

VELORA — Database Architecture Part 3: CRM, Finance, Settings

Module: CRM

customers

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
membership_tier_id	CHAR(36)	NULLABLE, FK→membership_tiers	
name	VARCHAR(150)	NOT NULL	
email	VARCHAR(150)	NULLABLE	
phone	VARCHAR(20)	NULLABLE	
gender	VARCHAR(10)	NULLABLE	Enum: male, female, other
birth_date	DATE	NULLABLE	
address	TEXT	NULLABLE	
city	VARCHAR(100)	NULLABLE	
member_code	VARCHAR(30)	NULLABLE	Membership card number
points_balance	INT UNSIGNED	DEFAULT 0	Current loyalty points
total_spent	BIGINT UNSIGNED	DEFAULT 0	Lifetime spending

Column	Type	Constraints	Description
total_transactions	INT UNSIGNED	DEFAULT 0	Lifetime transaction count
last_transaction_at	TIMESTAMP	NULLABLE	
notes	TEXT	NULLABLE	
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: (business_id, phone) , (business_id, member_code) , (business_id, email)

membership_tiers

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(50)	NOT NULL	e.g., "Silver", "Gold", "Platinum"
slug	VARCHAR(50)	NOT NULL	
min_spent	BIGINT UNSIGNED	DEFAULT 0	Min spending to qualify
min_transactions	INT UNSIGNED	DEFAULT 0	Min transactions
discount_percentage	DECIMAL(5,2)	DEFAULT 0	Auto discount for tier
points_multiplier	DECIMAL(5,2)	DEFAULT 1	Points earning multiplier
benefits	JSON	NULLABLE	List of benefits
color	VARCHAR(7)	NULLABLE	e.g., " #C0C0C0 "
sort_order	INT	DEFAULT 0	

Column	Type	Constraints	Description
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

loyalty_points

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
customer_id	CHAR(36)	FK→customers	
transaction_id	CHAR(36)	NULLABLE, FK→transactions	
type	VARCHAR(20)	NOT NULL	Enum: earned, redeemed, expired, adjusted, bonus
points	INT	NOT NULL	Positive=earn, Negative=spend
balance_after	INT UNSIGNED	NOT NULL	Balance after this entry
description	VARCHAR(255)	NULLABLE	
expires_at	TIMESTAMP	NULLABLE	
created_at	TIMESTAMP		

Indexes: (customer_id, created_at) , (business_id, expires_at)

Note

loyalty_points is a **ledger** (append-only). Current balance = customers.points_balance (denormalized for speed).

vouchers

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
code	VARCHAR(30)	NOT NULL	Voucher code
name	VARCHAR(100)	NOT NULL	
type	VARCHAR(20)	NOT NULL	Enum: percentage, fixed
value	DECIMAL(15,2)	NOT NULL	
min_purchase	BIGINT UNSIGNED	DEFAULT 0	
max_discount	BIGINT UNSIGNED	NULLABLE	
usage_limit	INT	NULLABLE	
usage_per_customer	INT	NULLABLE	
used_count	INT UNSIGNED	DEFAULT 0	
starts_at	TIMESTAMP	NULLABLE	
ends_at	TIMESTAMP	NULLABLE	
is_active	TINYINT(1)	DEFAULT 1	
applicable_tiers	JSON	NULLABLE	Restrict to membership tiers
applicable_products	JSON	NULLABLE	Restrict to products
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, code)

voucher_usages

Column	Type	Constraints	Description
id	CHAR(36)	PK	

Column	Type	Constraints	Description
voucher_id	CHAR(36)	FK→vouchers	
customer_id	CHAR(36)	NULLABLE, FK→customers	
transaction_id	CHAR(36)	FK→transactions	
discount_amount	BIGINT UNSIGNED	NOT NULL	
created_at	TIMESTAMP		

customer_debts

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
customer_id	CHAR(36)	FK→customers	
transaction_id	CHAR(36)	FK→transactions	
total_amount	BIGINT UNSIGNED	NOT NULL	
paid_amount	BIGINT UNSIGNED	DEFAULT 0	
remaining_amount	BIGINT UNSIGNED	GENERATED	
due_date	DATE	NULLABLE	
status	VARCHAR(20)	DEFAULT 'unpaid'	Enum: unpaid, partial, paid, overdue, written_off
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

customer_debt_payments

Column	Type	Constraints	Description
id	CHAR(36)	PK	
customer_debt_id	CHAR(36)	FK→customer_debts	
amount	BIGINT UNSIGNED	NOT NULL	
payment_method_id	CHAR(36)	FK→payment_methods	
reference	VARCHAR(100)	NULLABLE	
received_by	CHAR(36)	FK→users	
paid_at	TIMESTAMP	NOT NULL	
created_at	TIMESTAMP		

Module: Finance

expense_categories

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
parent_id	CHAR(36)	NULLABLE, FK→expense_categories	
name	VARCHAR(100)	NOT NULL	e.g., "Operasional", "Gaji", "Sewa"
slug	VARCHAR(100)	NOT NULL	
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

expenses

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	
category_id	CHAR(36)	FK→expense_categories	
expense_number	VARCHAR(50)	UNIQUE	
description	VARCHAR(500)	NOT NULL	
amount	BIGINT UNSIGNED	NOT NULL	
tax_amount	BIGINT UNSIGNED	DEFAULT 0	
total_amount	BIGINT UNSIGNED	NOT NULL	
expense_date	DATE	NOT NULL	
payment_method_id	CHAR(36)	NULLABLE, FK→payment_methods	
receipt_url	VARCHAR(500)	NULLABLE	Uploaded receipt
status	VARCHAR(20)	DEFAULT 'approved'	Enum: draft, pending, approved, rejected
approved_by	CHAR(36)	NULLABLE, FK→users	
created_by	CHAR(36)	FK→users	
note	TEXT	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

income

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	
income_number	VARCHAR(50)	UNIQUE	
source	VARCHAR(50)	NOT NULL	Enum: sales, other, interest, refund_received
description	VARCHAR(500)	NOT NULL	
amount	BIGINT UNSIGNED	NOT NULL	
income_date	DATE	NOT NULL	
reference_type	VARCHAR(100)	NULLABLE	transaction, etc.
reference_id	CHAR(36)	NULLABLE	
payment_method_id	CHAR(36)	NULLABLE	
created_by	CHAR(36)	FK→users	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

cash_flows

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	
type	VARCHAR(10)	NOT NULL	Enum: in, out

Column	Type	Constraints	Description
category	VARCHAR(30)	NOT NULL	Enum: sales, purchase, expense, debt_payment, debt_received, refund, other
amount	BIGINT UNSIGNED	NOT NULL	
balance_after	BIGINT	NOT NULL	Running balance
reference_type	VARCHAR(100)	NULLABLE	Source model
reference_id	CHAR(36)	NULLABLE	
description	VARCHAR(500)	NULLABLE	
flow_date	DATE	NOT NULL	
created_by	CHAR(36)	NULLABLE, FK→users	
created_at	TIMESTAMP		

Indexes: (business_id, branch_id, flow_date) , (business_id, type, flow_date)

Note

cash_flows is auto-populated via events. Every financial transaction creates a cash flow entry.

journals

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	NULLABLE	
journal_number	VARCHAR(50)	UNIQUE	
journal_date	DATE	NOT NULL	

Column	Type	Constraints	Description
description	VARCHAR(500)	NOT NULL	
reference_type	VARCHAR(100)	NULLABLE	
reference_id	CHAR(36)	NULLABLE	
total_debit	BIGINT UNSIGNED	DEFAULT 0	
total_credit	BIGINT UNSIGNED	DEFAULT 0	
is_auto	TINYINT(1)	DEFAULT 0	System-generated
status	VARCHAR(20)	DEFAULT 'posted'	Enum: draft, posted, voided
created_by	CHAR(36)	FK→users	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

journal_entries

Column	Type	Constraints	Description
id	CHAR(36)	PK	
journal_id	CHAR(36)	FK→journals	
account_code	VARCHAR(20)	NOT NULL	CoA code e.g., "1-1001"
account_name	VARCHAR(100)	NOT NULL	e.g., "Kas"
debit	BIGINT UNSIGNED	DEFAULT 0	
credit	BIGINT UNSIGNED	DEFAULT 0	
description	VARCHAR(255)	NULLABLE	
created_at	TIMESTAMP		

chart_of_accounts

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
parent_id	CHAR(36)	NULLABLE	
code	VARCHAR(20)	NOT NULL	e.g., "1-1001"
name	VARCHAR(100)	NOT NULL	
type	VARCHAR(20)	NOT NULL	Enum: asset, liability, equity, revenue, expense, cogs
is_system	TINYINT(1)	DEFAULT 0	
is_active	TINYINT(1)	DEFAULT 1	
balance	BIGINT	DEFAULT 0	Current balance
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, code)

Module: Settings

business_settings

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
group	VARCHAR(30)	NOT NULL	e.g., "pos", "tax", "invoice", "stock", "notification"
key	VARCHAR(50)	NOT NULL	

Column	Type	Constraints	Description
value	TEXT	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, group, key)

Default settings groups and keys:

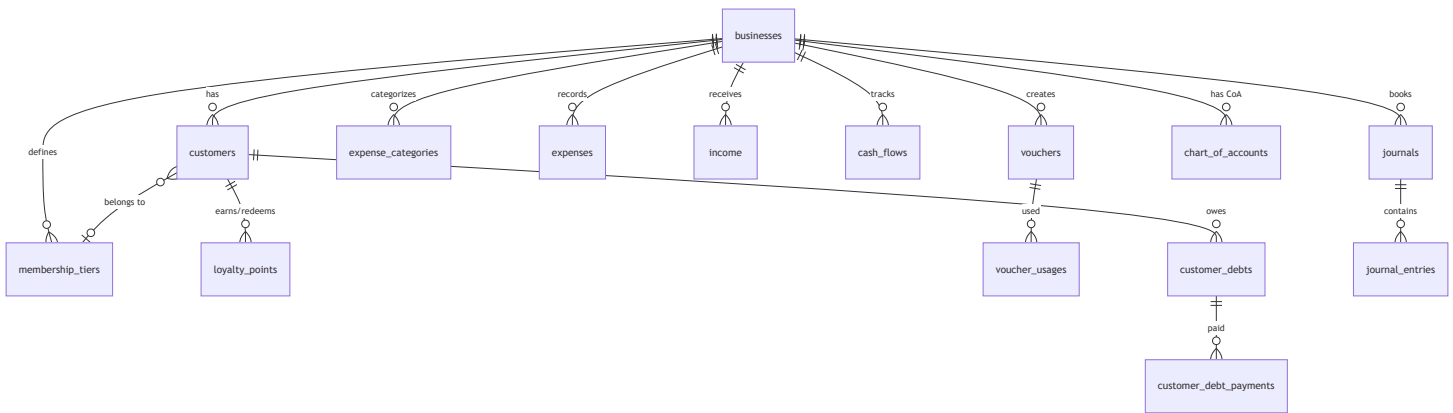
Group	Key	Default	Description
pos	receipt_header	""	Receipt top text
pos	receipt_footer	"Terima kasih!"	Receipt bottom text
pos	auto_print_receipt	"true"	
pos	allow_negative_stock	"false"	Sell below 0?
pos	default_tax_id	null	
pos	rounding_method	"none"	none, round_up, round_down, round_nearest
pos	rounding_precision	"100"	Round to nearest X
stock	costing_method	"avg"	avg, fifo
stock	low_stock_alert	"true"	
tax	default_tax_rate	"11"	PPN default
invoice	prefix	"INV"	
invoice	auto_number	"true"	
notification	low_stock_email	"true"	
notification	expiry_alert_days	"30"	Days before expiry

number_sequences

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
type	VARCHAR(30)	NOT NULL	transaction, purchase, refund, transfer, etc.
prefix	VARCHAR(10)	NOT NULL	e.g., "TRX", "PO"
separator	VARCHAR(5)	DEFAULT '-'	
include_date	TINYINT(1)	DEFAULT 1	Include YYYYMMDD
current_number	INT UNSIGNED	DEFAULT 0	Auto-increment
padding	INT	DEFAULT 5	Zero-pad length
reset_period	VARCHAR(10)	DEFAULT 'daily'	Enum: daily, monthly, yearly, never
last_reset_at	DATE	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, type)

Relationship Diagram — CRM + Finance



Master Table Count

Module	Tables	Names
Tenant	5	users, businesses, branches, business_user, branch_user
Subscription	5	subscription_plans, feature_limits, subscriptions, subscription_payments, subscription_logs
Auth/Security	5	roles, permissions, role_permission, activity_logs, audit_logs
POS	7	categories, brands, units, unit_conversions, products, product_variants, product_units, barcodes
Inventory	9	warehouses, product_warehouse, stock_movements, stock_transfers, stock_transfer_items, stock_opnames, stock_opname_items, stock_adjustments, stock_adjustment_items, batches
Sales	9	cashier_shifts, payment_methods, taxes, discounts, transactions, transaction_items, transaction_payments, refunds, refund_items
Purchasing	5	suppliers, purchases, purchase_items, supplier_debts, supplier_debt_payments
CRM	7	customers, membership_tiers, loyalty_points, vouchers, voucher_usages, customer_debts, customer_debt_payments
Finance	6	expense_categories, expenses, income, cash_flows, journals, journal_entries, chart_of_accounts
Settings	2	business_settings, number_sequences
TOTAL	~60	

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